DO NOW — RECALL

Packet switching, circuit switching and standards

Name: _____ Class: _____ Date: _____

1. Which TCP/IP layer contains TCP?

2. What does DNS usually provide when given a domain name?

3. Can a graph have more than one possible path between two vertices?

4. Name one advantage of sharing a network link instead of reserving it for one user.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Transport layer.

2. Model answer

An IP address for the name, when the appropriate DNS records exist.

3. Model answer

Yes.

4. Model answer

Capacity can be used by other traffic when a user is idle, improving efficiency.

Vocabulary — know these terms

packet switching — Data is split and forwarded across shared network links.

circuit switching — Communication uses a pre-established reserved path.

sequence information — Control information that allows ordered reconstruction.

retransmission — Sending missing or unacknowledged data again.

standard — Agreed specification supporting compatibility and interoperability.

NEW LEARNING

Read, annotate and explain

Split a message into packets

The sender can split data into separate packets or segments. Each includes control information such as addressing and/or sequence information at relevant protocol layers.

Packets may take different routes

Packet switching lets network traffic share links. IP forwarding can choose different paths; it does not guarantee the same arrival order.

Restore order when needed

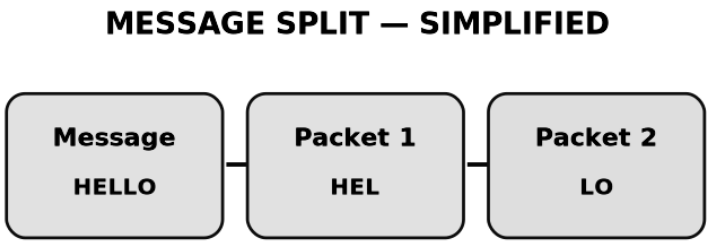
TCP provides a reliable ordered byte stream, using acknowledgements and retransmission if required. Do not confuse TCP guarantees with best-effort IP delivery.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

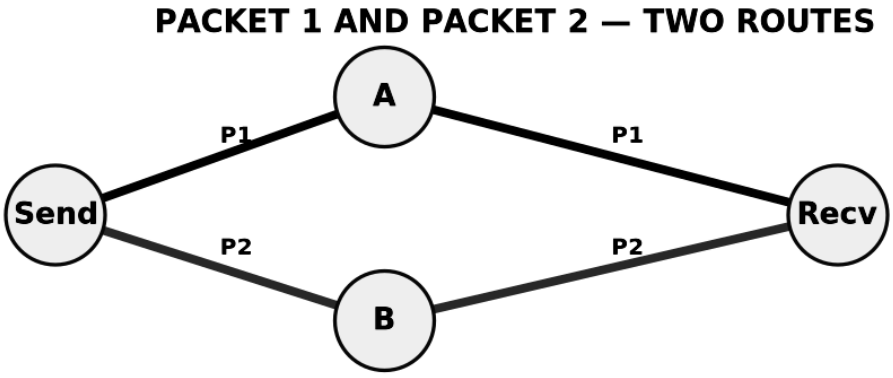
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



MESSAGE SPLIT — SIMPLIFIED. Message: HELLO; Packet 1: HEL; Packet 2: LO

New Learning

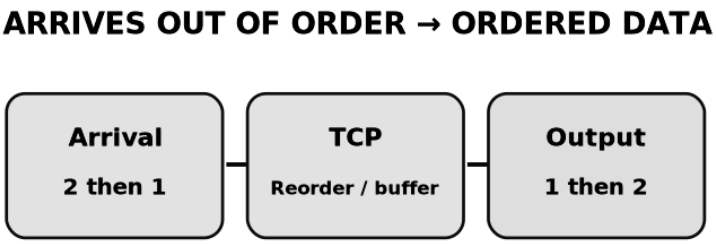


PACKET 1 AND PACKET 2 — TWO ROUTES. Packet 1 follows S to A to receiver; packet 2 follows S to B to receiver.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



ARRIVES OUT OF ORDER → ORDERED DATA. Arrival: 2 then 1; TCP: Reorder / buffer; Output: 1 then 2

I DO — WORKED EXAMPLE

Follow the worked method

Two TCP segments with sequence positions 1 and 2 travel on different routes and arrive 2, then 1. Explain the final data order.

Route

The network may forward them along different paths, so arrival order is not guaranteed.

Receive

TCP uses sequence information to buffer out-of-order data.

Deliver

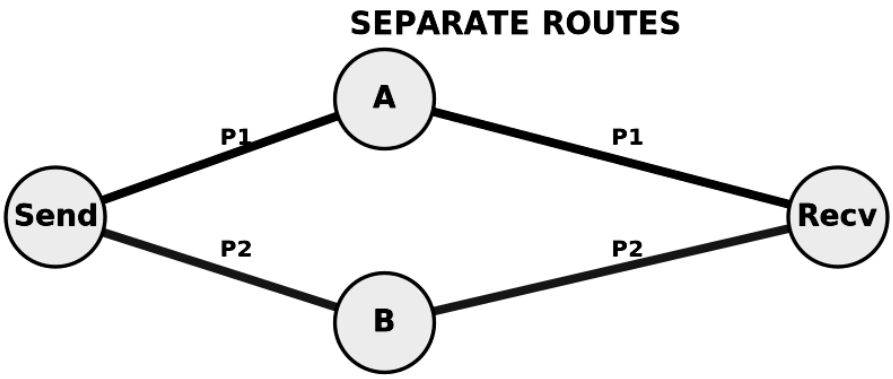
When missing earlier bytes arrive, TCP makes data available to the application in the correct order: first 1, then 2.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Two TCP segments with sequence positions 1 and 2 travel on different routes and arrive 2, then 1. Explain the

The network may forward them along different paths, so arrival order is not guaranteed.

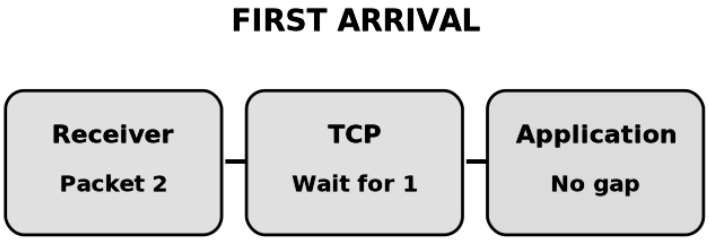


Different packets may traverse different paths

SEPARATE ROUTES. Packet 1 follows S to A to receiver; packet 2 follows S to B to receiver.

Two TCP segments with sequence positions 1 and 2 travel on different routes and arrive 2, then 1. Explain the

The network may forward them along different paths, so arrival order is not guaranteed.



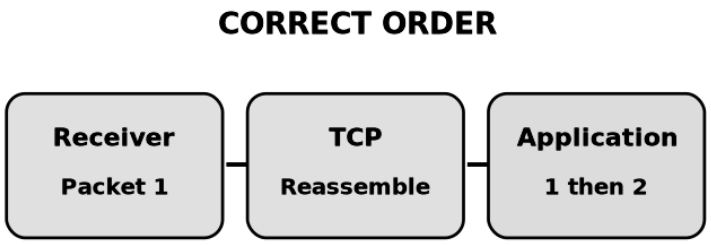
FIRST ARRIVAL. Receiver: Packet 2; TCP: Wait for 1; Application: No gap

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Two TCP segments with sequence positions 1 and 2 travel on different routes and arrive 2, then 1. Explain the

The network may forward them along different paths, so arrival order is not guaranteed.



CORRECT ORDER. Receiver: Packet 1; TCP: Reassemble; Application: 1 then 2

WE DO — GUIDED PRACTICE

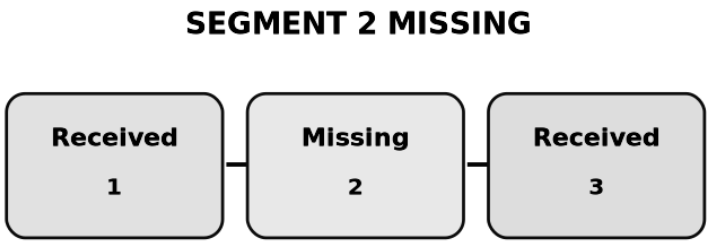
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Three TCP data segments arrive as 1, 3, with segment 2 missing. Explain what TCP can do.

The receiver recognises that data between segment 1 and segment 3 is missing.



SEGMENT 2 MISSING. Received: 1; Missing: 2; Received: 3

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Detect gap The receiver recognises that data between segment 1 and segment 3 is missing.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Give one difference between packet switching and circuit switching. [2]

Question 2

Data labelled 1,2,3 arrives as 2,1,3 using TCP. Explain how TCP handles the out-of-order arrival and what the application receives. [4]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain one advantage of packet switching and why communication standards are important for different manufacturers' devices. [4]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

IP guarantees that packets always arrive in the order sent.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

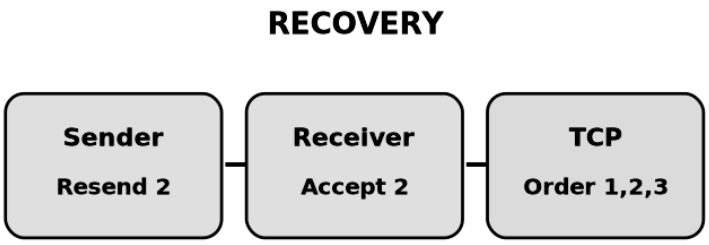
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Three TCP data segments arrive as 1, 3, with segment 2 missing. Explain what TCP can do.

The receiver recognises that data between segment 1 and segment 3 is missing.



RECOVERY. Sender: Resend 2; Receiver: Accept 2; TCP: Order 1,2,3

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Recover

TCP can cause the missing data to be retransmitted; once the gap is filled, data is delivered in order.

IMPROVE: Explain one correction to your own reasoning.